# ODC AI Mentor Functional Requirements Document (PRD)

**Project:** Seoul Station Master Smart Application **Target:** Automated Generation via ODC AI Mentor

> **ODC AI Mentor Guide:** This document specifies requirements for a web/mobile application based on the public dataset "서울시 역사마스터 정보.csv". It defines OutSystems Data Cloud (ODC) Entities, Reactive Web UI screens, Server Actions, REST APIs, and AI Chatbot integration.

## 1. System Overview & Objective

This system manages Seoul Metro Station Master data (station name, line, station ID, GPS coordinates, road/lot addresses). It enables users to search nearby stations and interact via an AI-driven conversational interface to retrieve station and subway line information efficiently.

## 2. ODC Entity Data Model Architecture

Definition of ODC Database Entities to be created based on the input CSV dataset.

| Entity Name | Attribute Name | Data Type | Mandatory | Description & CSV Mapping |
|---|---|---|---|---|
| **StationMaster** (Main Table) | Id | Identifier (Long) | Yes (PK) | Auto-generated primary key |
| | StationId | Text (50) | Yes | Station Unique Identifier (CSV: `역사ID`) |
| | StationName | Text (100) | Yes | Station Name (CSV: `역명`) |
| | LineName | Text (50) | Yes | Subway Line Name (CSV: `노선명`) |
| | RoadAddress | Text (255) | No | Road Address (CSV: `역사도로명주소`) |
| | JibunAddress | Text (255) | No | Lot Number Address (CSV: `역사위치주소`) |
| | Latitude | Decimal (18, 8) | No | GPS Latitude (Y Coordinate) |
| | Longitude | Decimal (18, 8) | No | GPS Longitude (X Coordinate) |
| **StationBookmark** (User Favs) | Id | Identifier (Long) | Yes (PK) | Primary key |
| | UserId | User Identifier | Yes (FK) | Authenticated User ID |
| | StationId | Identifier (Long) | Yes (FK) | Reference to StationMaster.Id |

## 3. Screen Specifications & UI Requirements

| Screen Name | UI Components | Logic & Workflows |
| --- | --- | --- |
| **StationList**<br>(Main Search) | - Search Input Bar<br>- Line Dropdown Filter<br>- Table / Grid View<br>- Pagination Controls | - Real-time filtering by Station Name or Road Address<br>- Add Aggregate filter condition on line selection (`LineName = SelectedLine`)<br>- Row click navigates to `StationDetail` |
| **StationDetail**<br>(Detail View) | - Station Info Card<br>- Address Section<br>- Bookmark Toggle Button<br>- Map View Container | - Fetch single station details using URL parameter `StationId`<br>- Trigger `ToggleBookmark` Server Action on button click<br>- Render interactive map pin using GPS coordinates |
| **StationImportCSV**<br>(Data Upload) | - File Upload Widget<br>- Import Progress Bar<br>- Result Summary Card | - Execute `BulkUploadStationMaster` Server Action upon upload<br>- Perform Upsert operation validating duplicate `StationId` |
| **AIAssistantChat**<br>(AI Chatbot) | - Chat History Container<br>- Prompt Input Box<br>- Quick Action Badges | - Process natural language queries (e.g., "Find stations near Gangnam on Line 2")<br>- Invoke AI Agent Action to query DB and return formatted responses |

## 4. Core Business Logic & Server Actions

- **GetStationsByFilter:** Filters the `StationMaster` Aggregate using `SearchKeyword` and `LineFilter`, returning paginated station records.
- **BulkUploadStationMaster:** Parses the uploaded `.csv` binary stream into list structures and sequentially executes `CreateOrUpdateStationMaster`.
- **GetNearestStation:** Accepts current user GPS coordinates (Latitude/Longitude) and applies the Haversine formula to return the 3 nearest stations.

## 5. ODC AI Mentor Prompt Instructions

```
"Create a Reactive Web Application for Seoul Station Master Management. Integrate the StationMaster entity with StationId, StationName, LineName, RoadAddress, JibunAddress, and set Latitude and Longitude to Decimal (precision 8). Generate a dashboard listing stations with search filters, a detailed station view, a CSV import module for '서울시 역사마스터 정보.csv', and an AI assistant chat interface."
```